

Updated: May 21, 2026

Guidelines for the final project

The final project report should be submitted as a **single Jupyter Notebook file**, and run so that it **contains also all the output** (figures / output text).

The deadline for submission will be communicated.

Evaluation criteria:

(1) Implementation of the relevant learned analysis modules for addressing the project's questions.

For each individual implemented analysis module, the following aspects will be assessed:

- Clarity of stating (or short describing) the driving motivation / **question** for the analysis (should be done in a Markdown cell before the cell of the analysis code)
- Correctness of the implementation, clarity of the **code**, and code comments when needed. (the analysis code cell)
- Clarity and completeness of the generated **figures** (which should appear below the analysis code cell): Axis titles, legends, etc.
- Clarity and depth of the **interpretation** of the obtained results (should be done in a markdown cell after the cell with the analysis code and the generated figure).

(2) Clear structuring and logical organization of the whole Jupyter Notebook:

The whole Notebook, once run as a whole from the beginning until the end, should generate the correct results and be organized with sections and subsections (when applicable), including the corresponding titles in a logical manner.

Analysis guidelines for the project:

Specific project-related tasks and general guidelines will be provided by me gradually, in parallel with the learning of the corresponding topics in the sessions. This information will be gradually appended to this document below (which should therefore be referred to as a dynamic document).

Submission deadline: 15.06.2026

Project task 1: Data import and organization

Import the data of the project and process it as described in “DataAnalyticsPython_Exercises_04.pdf”.

Thus, you should at the end derive the `df_diabetes_metadata` and `df_diabetes_features` as Excel files named `Table_diabetes_metadata.xlsx` and `df_diabetes_features.xlsx`, respectively.

Organize this part in a clean manner, at the beginning of your Jupyter Notebook report file.

Project task 2: Visualizing of the data

(1) Make box-plots to visualize and compare the distribution of the raw (!) values of all the features in `df_diabetes_features`.

- Use the exercises that were done in DataAnalytics “Python_Exercises_05.pdf” as a reference.
- What is the motivation to do these plots? does it make sense to this visualization also on the data after standardization?

(2) Make box-plot comparing the Age of the women and men participants in the dataset.

(3) Make a scatter plot for each possible 2-feature pair (including all the numeric features: so including all the features in `df_diabetes_features` as well as Age).

(4) What can you conclude at this stage from visual inspection of the plots?

Project task 3: Testing for pair-wise associations between features

Apply the proper statistical tests to check for pair-wise association between features:

(1) Are there features (including Age) in the dataset which differ significantly between women and men?

(2) Are there features (including Age) in the dataset which have significant correlation between them? Consider which correlation test is proper to use here

Project task 4: Evaluating of there are clusters in the data using hierarchical clustering

(1) Generate a copy of `df_diabetes_features`, name it `df_diabetes_basal_features` and omit from `df_diabetes_basal_features` the “Progression” feature.

So, `df_diabetes_basal_features` should contain the following 8 features: 'BMI', 'BP', 'TC', 'LDL', 'HDL', 'TCH', 'LTG', 'Glu'

(2) Using hierarchical cluster analysis, assess if there are clear clusters in the data (`df_diabetes_basal_features`).

Important: How do you need to further process `df_diabetes_basal_features`, so that applying cluster analysis on it will be meaningful?

Generate dendrograms to visualize the clustering results. Try different linkage methods (“average”, “ward” and possibly others). Interpret the results.

(3) The `fcluster` function (import with: `from scipy.cluster.hierarchy import fcluster`) enables to derive from a given linkage matrix and threshold an assignment of each of the data point to a given cluster.

Use this, in order to assess if the clusters that are obtained (choose one of what you have tried in (2)) are associated with the following features:

- Gender
- Progression (categorized to two: (i) above the median progression and (ii) below or equal the median progression)
- Age (categorized to two: (i) above the median age and (ii) below or equal the median age)

Project task 5: Exploring multidimensional patterns with PCA

(1) Process `df_diabetes_basal_features` for PCA analysis. First check if it is possible to apply log-transformation, and if yes then apply it. Next apply data standardization.

(2) Apply the PCA on the data you obtained in (1). Generate then a plot of PC1 vs PC2, indicating on the axes the percentages of explained variance. Generate an additional similar plot PC3 vs PC4. Interpret the results.

(3) Evaluate and visualize the loading of the original features on PC1 and PC2. What can you learn from the results?

(4) Generate the following additional plots:

- a. As in (2), but color-code the points according to Gender
- b. As in (2), but color-code the points according to Progression (categorized to two: (i) above the median progression and (ii) below or equal the median progression)
- c. As in (2), but color-code the points according to Age (categorized to two: (i) above the median age and (ii) below or equal the median age)

Interpret the results.

Project task 6: Applying multiple linear regression

Apply multiple linear regression for predicting the “Progression” feature based on the 8 features of the `df_diabetes_basal_features` ('BMI', 'BP', 'TC', 'LDL', 'HDL', 'TCH', 'LTG', 'Glu'). General guidelines:

- Partition the data to train set and test set
- Perform data standardization of the two sets (make sure to do that in a manner which is right for each of the two sets)
- Perform the linear regression analysis, and evaluate the fit

- Assess the importance of the features for the prediction of the “Progression” feature
- Interpret the results

Project task 7: Applying logistic regression

Apply logistic regression for predicting “High progression” versus “Low progression” based on the 8 features of the `df_diabetes_basal_features` ('BMI', 'BP', 'TC', 'LDL', 'HDL', 'TCH', 'LTG', 'Glu').

“High progression”: the “Progression” feature value is $>$ “Progression” feature median

“Low progression”: the “Progression” feature value is $\leq$ “Progression” feature median

General guidelines:

- Create a feature “Progression_class”, corresponding to the “Progression” feature, but with the value 0 for Progression values which are $\leq$ Progression feature median, and value 1 for Progression values which are $>$ Progression feature median
- Partition the data to train set and test set
- Perform data standardization to the two sets (make sure to do that in a manner which is right for each of the two sets)
- Perform the logistic regression analysis with Elastic net regularization
- Generate the ROC curve and derive also the area under the ROC curve (ROC AUC)
- Assess the important of the features for the prediction of the “Progression_class” feature
- Interpret the results